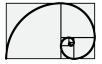


Three Phase Inductors

Mathematical Model and Construction

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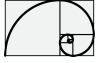
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1 Three phase inductance: mathematical model

Three-phase inductance with four limbs:

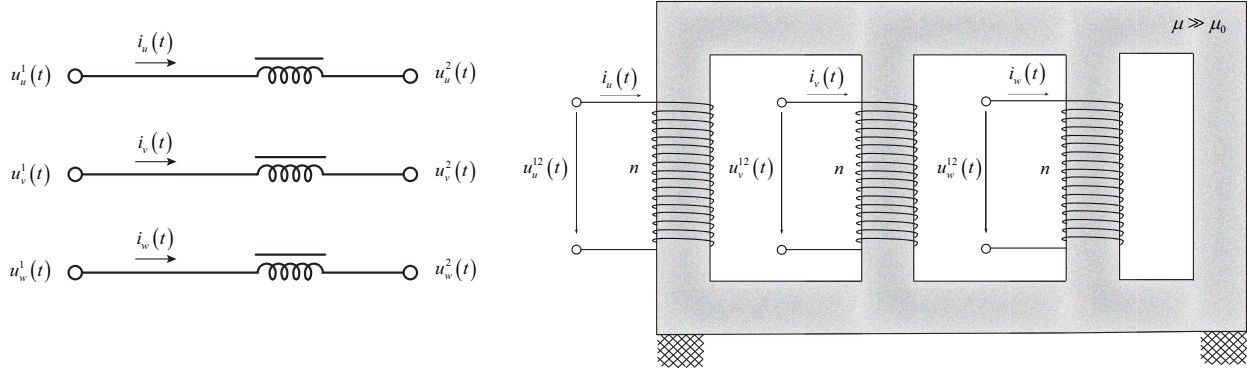


Figure 1: Three-phase inductance: differential and common operational mode with four magnetic limbs.

Three-phase inductance with three limbs - pure differential mode:

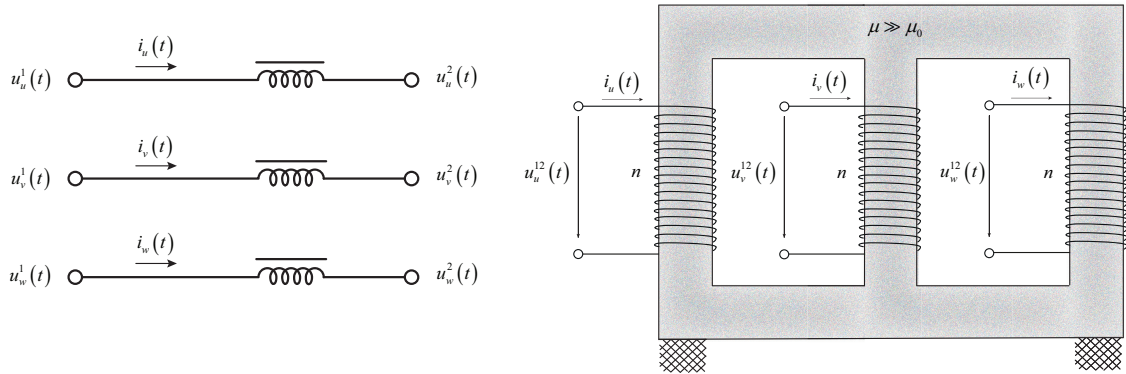


Figure 2: Three-phase inductance: pure differential operative mode with three magnetic limbs.

Three-phase inductance with one limb - pure common mode:

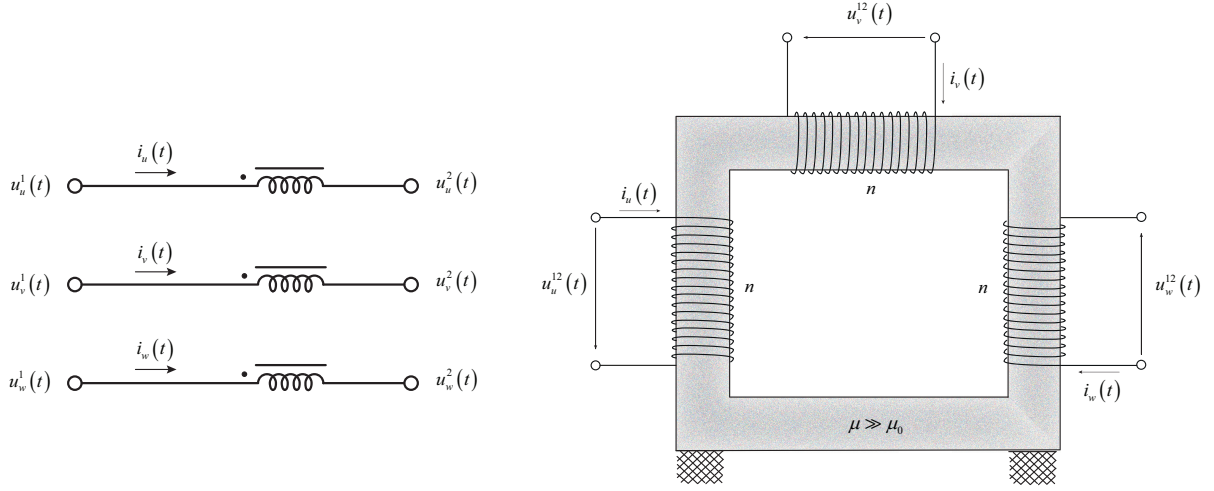
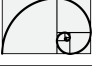


Figure 3: Three-phase inductance: pure common operative mode with one magnetic limbs (classical toroidal core is used for low power application).

Mathematical model:

$$\begin{cases} u_u^{12}(t) = u_u^{cm}(t) + u_u^{dm}(t) \\ u_v^{12}(t) = u_v^{cm}(t) + u_v^{dm}(t) \\ u_w^{12}(t) = u_w^{cm}(t) + u_w^{dm}(t) \end{cases} \quad (1.1)$$

where

$$\begin{cases} u_u^{dm}(t) + u_v^{dm}(t) + u_w^{dm}(t) = 0 \\ \frac{1}{3} \left(u_u^{cm}(t) + u_v^{cm}(t) + u_w^{cm}(t) \right) = u_{cm}(t) \end{cases} \quad (1.2)$$

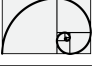
$$\begin{cases} u_u^{12}(t) - Ri_u(t) - L_a \frac{di_u}{dt}(t) + L_m \frac{di_v}{dt}(t) + L_m \frac{di_w}{dt}(t) = 0 \\ u_v^{12}(t) - Ri_v(t) - L_a \frac{di_v}{dt}(t) + L_m \frac{di_u}{dt}(t) + L_m \frac{di_w}{dt}(t) = 0 \\ u_w^{12}(t) - Ri_w(t) - L_a \frac{di_w}{dt}(t) + L_m \frac{di_u}{dt}(t) + L_m \frac{di_v}{dt}(t) = 0 \end{cases} \quad (1.3)$$

which results in matrix form

$$\vec{u}_{uvw}^{12}(t) - \mathbf{R} \vec{i}_{uvw}(t) - \mathbf{L} \frac{d}{dt} \vec{i}_{uvw}(t) = 0 \quad (1.4)$$

where

$$\mathbf{R} = \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} L_a & -L_m & -L_m \\ -L_m & L_a & -L_m \\ -L_m & -L_m & L_a \end{bmatrix}. \quad (1.5)$$



in order to derive the equivalent *differential* and *common* mode equations the following transformation matrix is accounted

$$\mathbf{T} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}. \quad (1.6)$$

where

$$\vec{u}_{uvw}^{12} = \begin{bmatrix} u_u^{12}(t) \\ u_v^{12}(t) \\ u_w^{12}(t) \end{bmatrix} = \mathbf{T}^{-1} \begin{bmatrix} u_{uv}^{dm}(t) \\ u_{vw}^{dm}(t) \\ u_{wu}^{dm}(t) \\ u_{cm}(t) \end{bmatrix}. \quad (1.7)$$

where

$$\begin{bmatrix} u_{uv}^{dm}(t) \\ u_{vw}^{dm}(t) \\ u_{wu}^{dm}(t) \\ u_{cm}(t) \end{bmatrix} - \begin{bmatrix} R & -R & 0 \\ 0 & R & -R \\ -R & 0 & R \\ R & R & R \end{bmatrix} \begin{bmatrix} i_u(t) \\ i_v(t) \\ i_w(t) \end{bmatrix} - \begin{bmatrix} L_a + L_m & -(L_a + L_m) & 0 \\ 0 & L_a + L_m & -(L_a + L_m) \\ -(L_a + L_m) & 0 & L_a + L_m \\ L_a - 2L_m & L_a - 2L_m & L_a - 2L_m \end{bmatrix} \begin{bmatrix} \frac{di_u}{dt}(t) \\ \frac{di_v}{dt}(t) \\ \frac{di_w}{dt}(t) \end{bmatrix} = 0 \quad (1.8)$$

expressing in terms of $\begin{bmatrix} u_{dmu}^{12}(t) & u_{dmv}^{12}(t) & u_{dmw}^{12}(t) \end{bmatrix}^T$ and $u_{cm}(t)$ we obtain the following system

$$\vec{u}_{duvw}^{12}(t) - \mathbf{R}\vec{i}_{uvw}(t) - \mathbf{L}_{dm}\frac{d}{dt}\vec{i}_{uvw}(t) = 0 \quad (1.9)$$

where

$$\mathbf{R} = \begin{bmatrix} R & 0 & 0 \\ 0 & R & 0 \\ 0 & 0 & R \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} L_a + L_m & 0 & 0 \\ 0 & L_a + L_m & 0 \\ 0 & 0 & L_a + L_m \end{bmatrix}. \quad (1.10)$$

and

$$u_{cm}(t) - Ri_{cm}(t) - (L_a - 2L_m)\frac{d}{dt}i_{cm}(t) = 0 \quad (1.11)$$

where $i_{cm}(t) = \frac{1}{3}(i_u(t) + i_v(t) + i_w(t))$. Hence, in a three-phase inductance we have two inductive contributions

- differential mode (also called *positive sequence inductance*) given by $L_p = L_a + L_m$;
- common mode (also called *zero sequence inductance*) given by $L_0 = L_a - 2L_m$

where L_a is the self-inductance, and L_m is the mutual-inductance.



The relation between sequence values and machine values can be summarized as follows

$$\begin{cases} L_p = L_a + L_m \\ L_0 = L_a - 2L_m \\ L_m = \frac{1}{3}(L_p - L_0) \\ L_a = \frac{1}{2}(L_p + L_0 + L_m) \end{cases} \quad (1.12)$$

Commercially speaking three-phase inductor can be either *common mode* or *differential mode*:

- Typically three-phase three-limbs inductors present only differential inductance. The residual common mode inductance is associated to the leakage inductance, see Figure 2.
- Typically three-phase common mode inductors are built around a common single limb core. The residual differential mode inductance is associated to the leakage inductance, see Figure 3.

2 Saturation effects

The saturation effects can be modeled assuming L_a and L_m as function of the current, as follows

$$\begin{cases} L_{au} = L_a^{nom} k_{sat}^a(i_u) + L_{a0} \rightarrow L_{a0} & \text{for } i \gg i_{sat} \\ L_{av} = L_a^{nom} k_{sat}^a(i_v) + L_{a0} \rightarrow L_{a0} & \text{for } i \gg i_{sat} \\ L_{aw} = L_a^{nom} k_{sat}^a(i_w) + L_{a0} \rightarrow L_{a0} & \text{for } i \gg i_{sat} \\ L_{mu} = L_m^{nom} k_{sat}^m(i_u) + L_{m0} \rightarrow L_{m0} & \text{for } i \gg i_{sat} \\ L_{mv} = L_m^{nom} k_{sat}^m(i_v) + L_{m0} \rightarrow L_{m0} & \text{for } i \gg i_{sat} \\ L_{mw} = L_m^{nom} k_{sat}^m(i_w) + L_{m0} \rightarrow L_{m0} & \text{for } i \gg i_{sat} \end{cases} \quad (2.1)$$

where L_{a0} and L_{m0} are the equivalent self and mutual inductance in air, condition which raises e.g. during short circuit current.

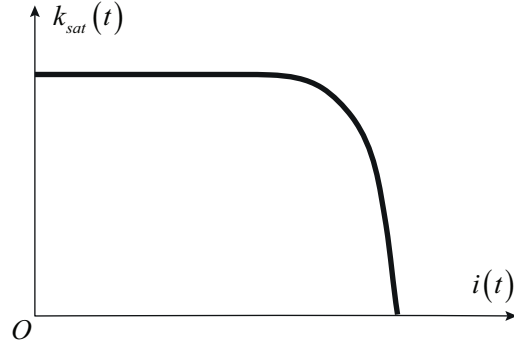
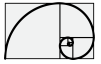


Figure 4: Saturation curve applied to L_a and L_m , self and mutual inductance respectively.

$$\mathbf{L}(\vec{i}_{uvw}) = \begin{bmatrix} L_{ls} + L_a^{nom} k_{sat}^a(i_u) + L_{a0} & -[L_m^{nom} k_{sat}^m(i_v) + L_{m0}] & -[L_m^{nom} k_{sat}^m(i_w) + L_{m0}] \\ -[L_m^{nom} k_{sat}^m(i_u) + L_{m0}] & L_{ls} + L_a^{nom} k_{sat}^a(i_v) + L_{a0} & -[L_m^{nom} k_{sat}^m(i_w) + L_{m0}] \\ -[L_m^{nom} k_{sat}^m(i_u) + L_{m0}] & -[L_m^{nom} k_{sat}^m(i_v) + L_{m0}] & L_{ls} + L_a^{nom} k_{sat}^a(i_w) + L_{a0} \end{bmatrix} \quad (2.2)$$



References

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- [2] W. G. Hurley, W. H. Woelfle - *Transformers and Inductors for Power Electronics. Theory design and applications*. J. Wiley 2013.